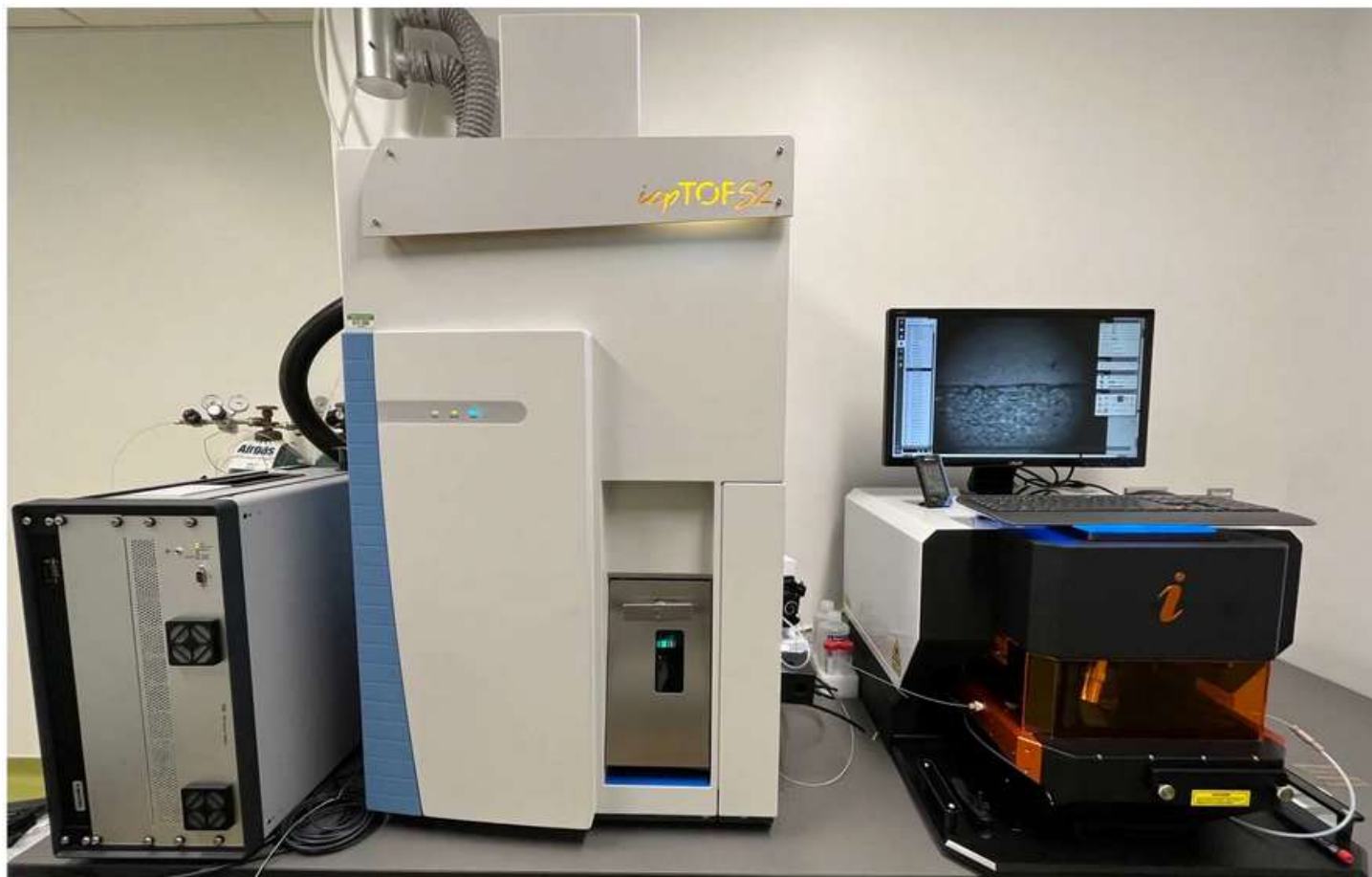


Bio Image 266 Laser Ablation System TofWerk S2 ICP-TOF-MS



QUANTITATIVE BIO ELEMENT ANALYSIS AND MAPPING CENTER

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Pre-start checklist

Turn on the chiller

The chiller is located just outside the lab. Press the power button to turn on the chiller.



Turn on the argon flow

The argon tanks are next to the chillers.



Check the tank gauge to be sure there is enough argon in the tank. If the tank is nearly empty, contact Keith MacRenaris (macrenar@msu.edu).



Make sure the correct regulator is attached to the argon tank so gas can flow to the instrument. The tubing should have tape labeled "ICP TOF MS" on it. If the correct regulator is attached, check the valve on the tank nozzle that is attached to the regulator by twisting counterclockwise. If it is fully open, it should give resistance when twisted. When fully opened, rotate back a half twist.



Turn on the helium

The helium tanks are located to the left of the instrument.



Open the tank by twisting counterclockwise. If the regulator does not read roughly 20 PSI, the tank must be replaced.



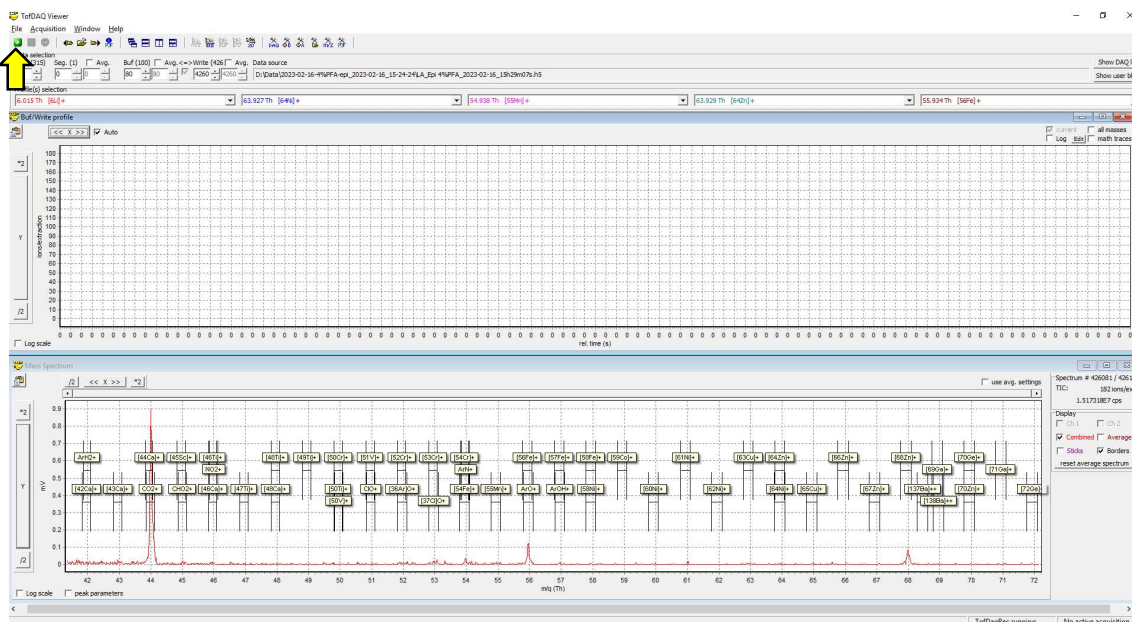
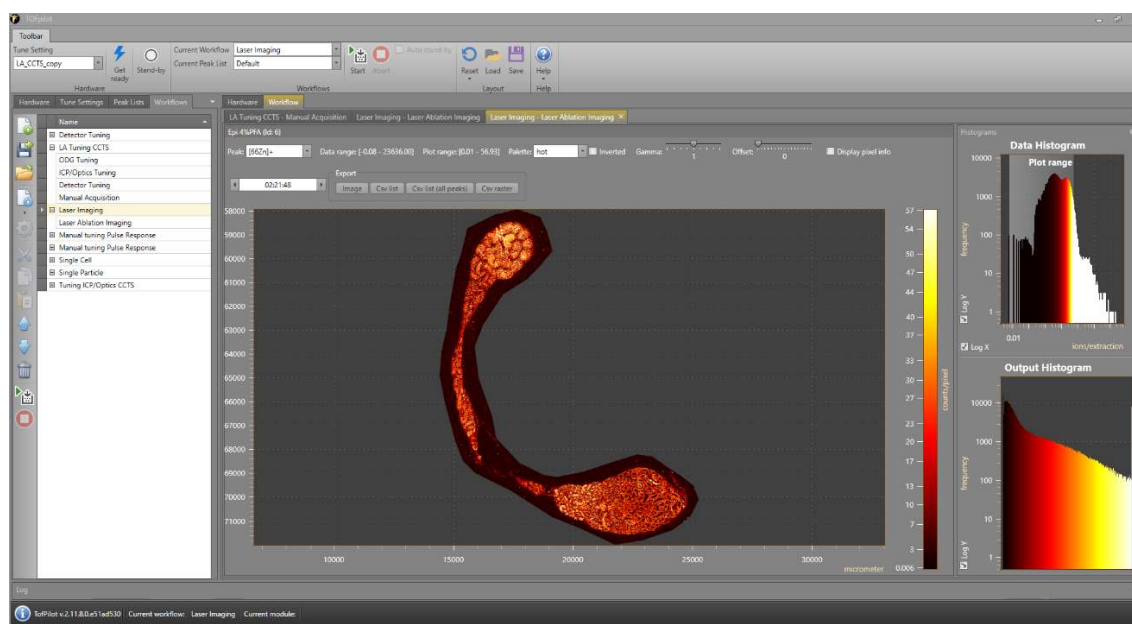
Starting the instrument

Opening the software

Select TofPilot from the desktop.



The program will open two separate windows – TofPilot and TofDAQ Viewer. Note: If the play button in TofDAQ Viewer remains grayed out, restart the program.

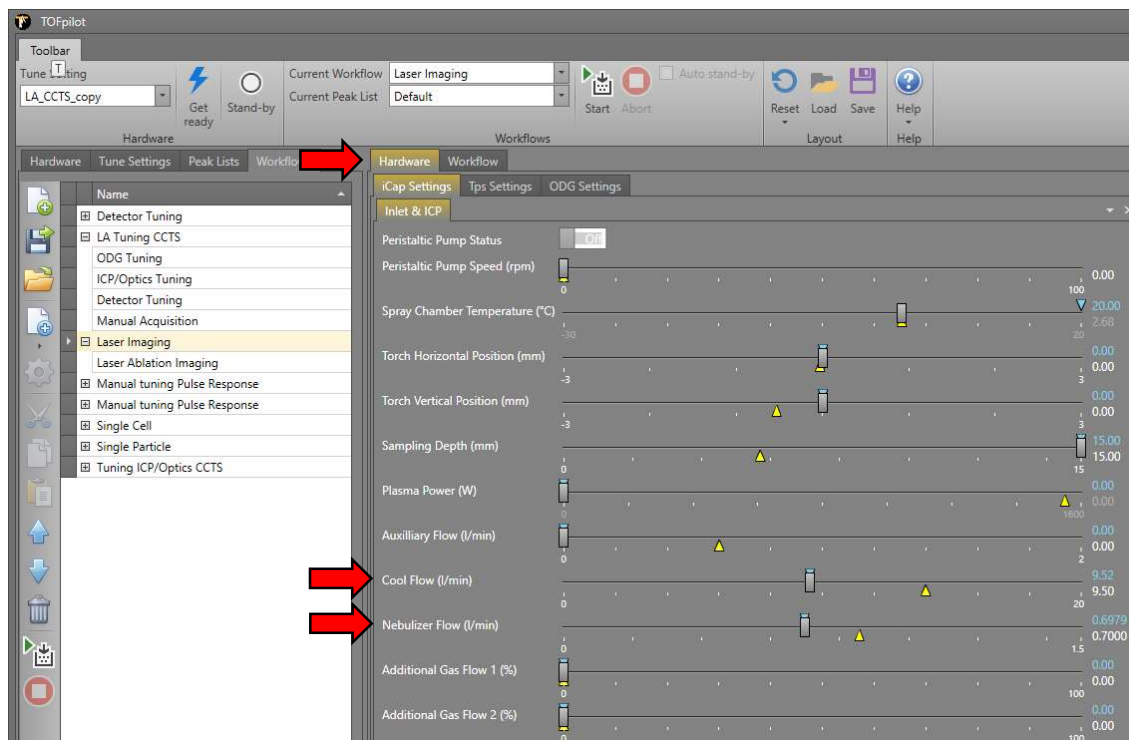


On the laser computer, the software will likely already be open. If not, select ActiveView2.



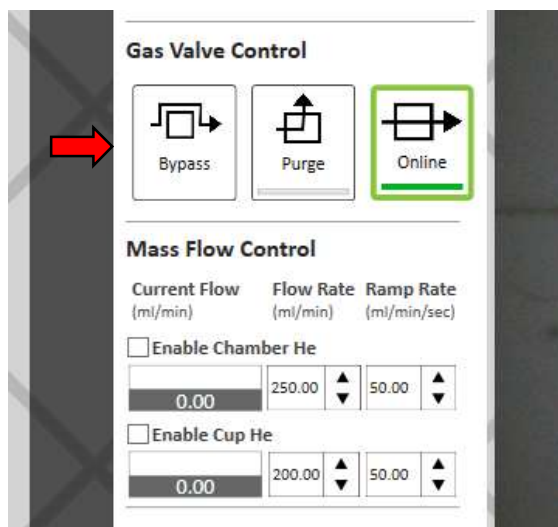
Preparing the torch for ignition

In the hardware tab, move the Nebulizer Flow and Cool Flow gauges about halfway. While loading the sample, this will allow air to be sufficiently flushed from the system.



Loading a sample

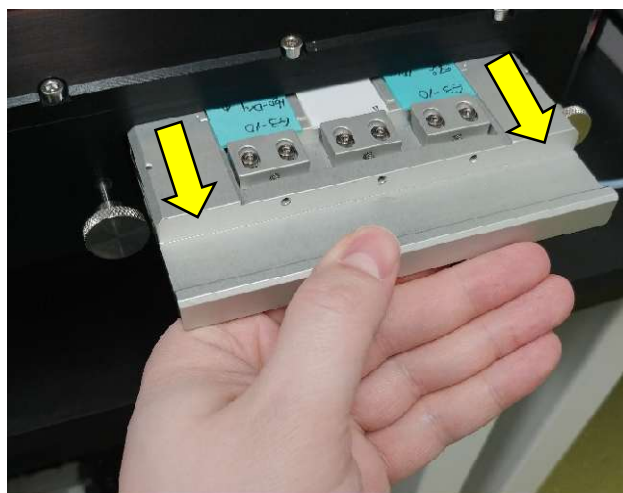
Under the laser software, click the bypass button. Wait until the reticule in the laser software moves to the upper left corner (marked by a red box) before proceeding.



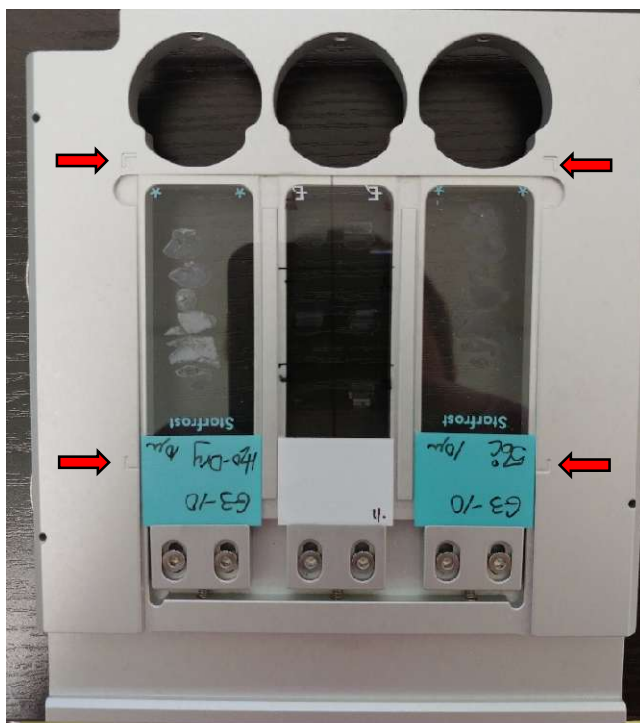
Slightly unscrew the sample holder block and lift off the block.



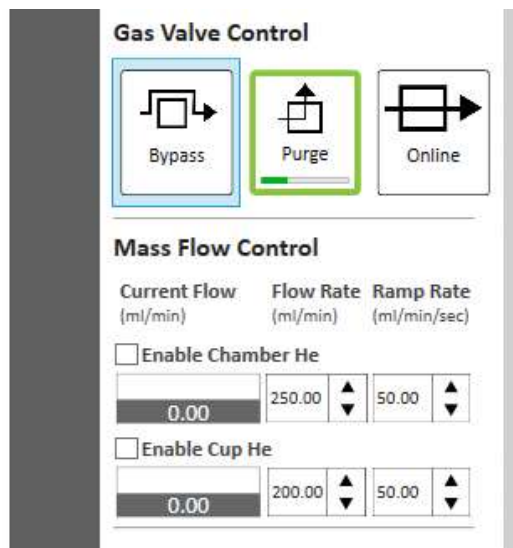
Remove the sample holder and place the appropriate slides into the holder.



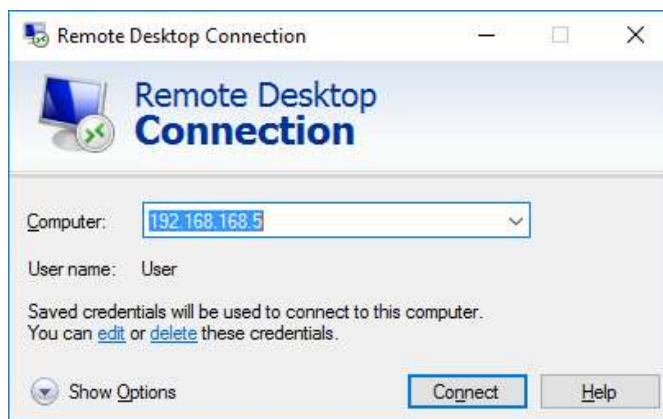
Take a picture of the slides in the sample holder. Note: The picture should be taken such that the camera is parallel to the sample holder to prevent stretching of the image. Also ensure that the fiducials (right angle notches on the corners of the slide holder) are in the picture.



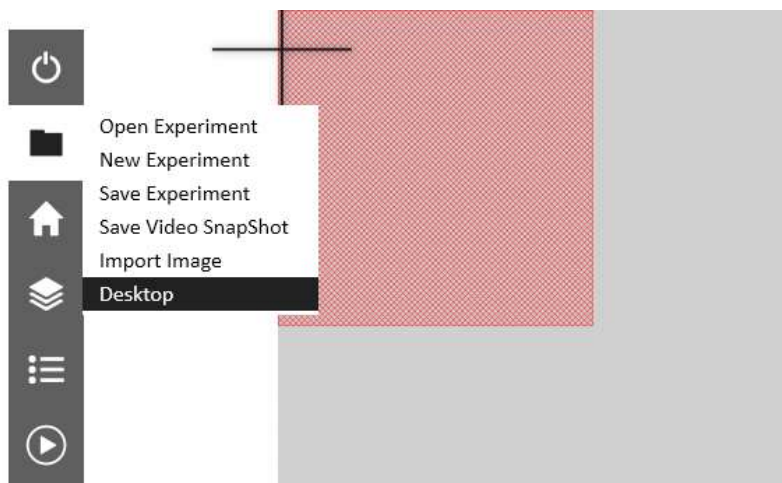
Place the sample holder back into the laser, put the block back and finger tighten the screws. Select Purge under Gas Valve Controls.



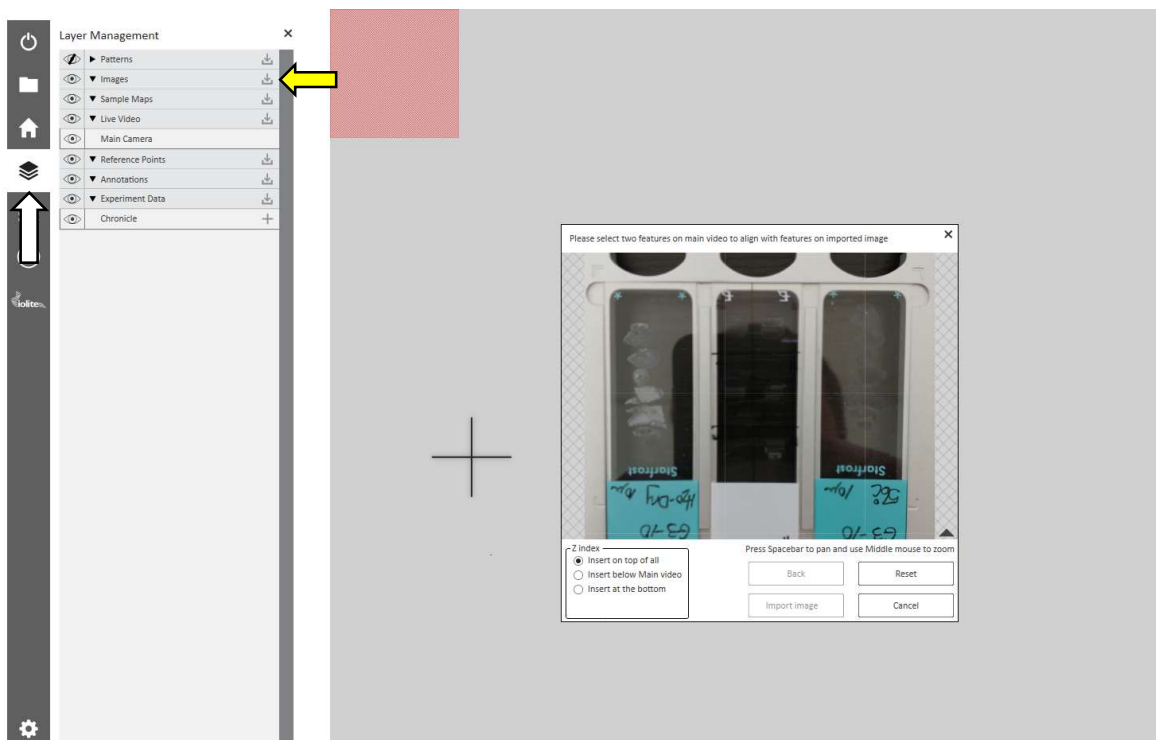
While the chamber is purging, email the sample picture to yourself and download it. Once the purge has completed (otherwise the software may crash), on the ICP computer, open “Remote Desktop Connection” and click Connect (the IP address should already be input). This will open the laser desktop on the ICP computer.



Click the folder icon then click desktop in the laser software to minimize the laser software. Copy-paste the picture from the downloads folder on the ICP computer to the pictures folder on the laser computer. After transferring the image to the laser computer, the remote desktop can be closed.



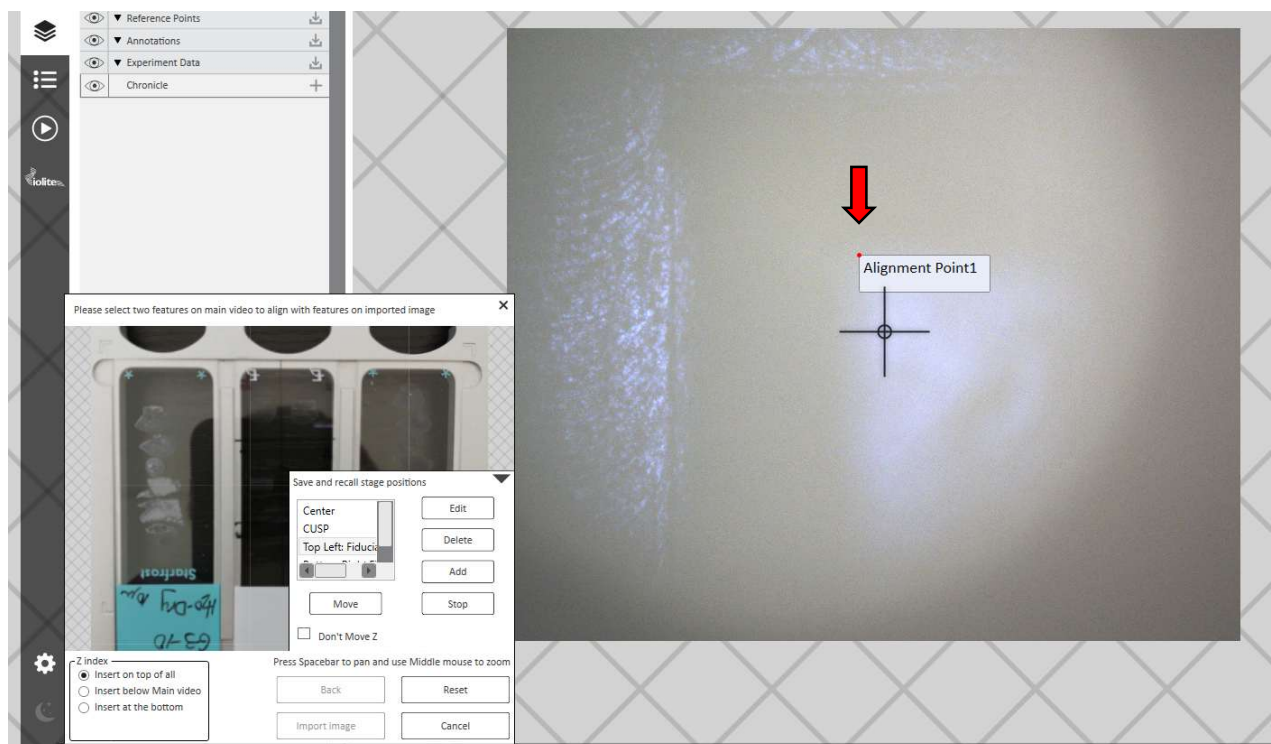
To load the image into the laser software, go to the Layer Management tab. Click the upload button in the Images section and select the image file. After loading the image, a small window will pop up to assign the top left and bottom right fiducials.



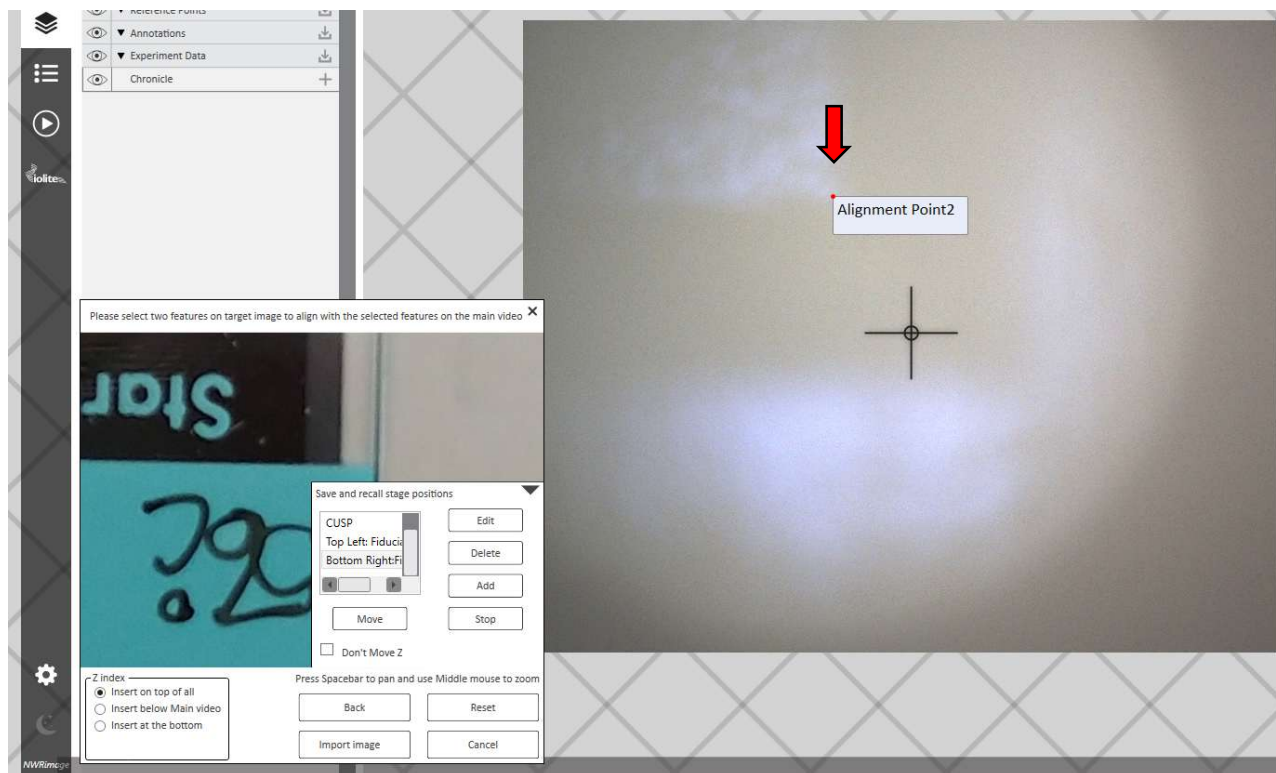
Click the small arrow in the bottom right of the image to bring up a menu. This will set the fiducial positions in the instrument. Click the "Top Left" fiducial and click the Move button to move the stage (the position should already be roughly near the fiducial).



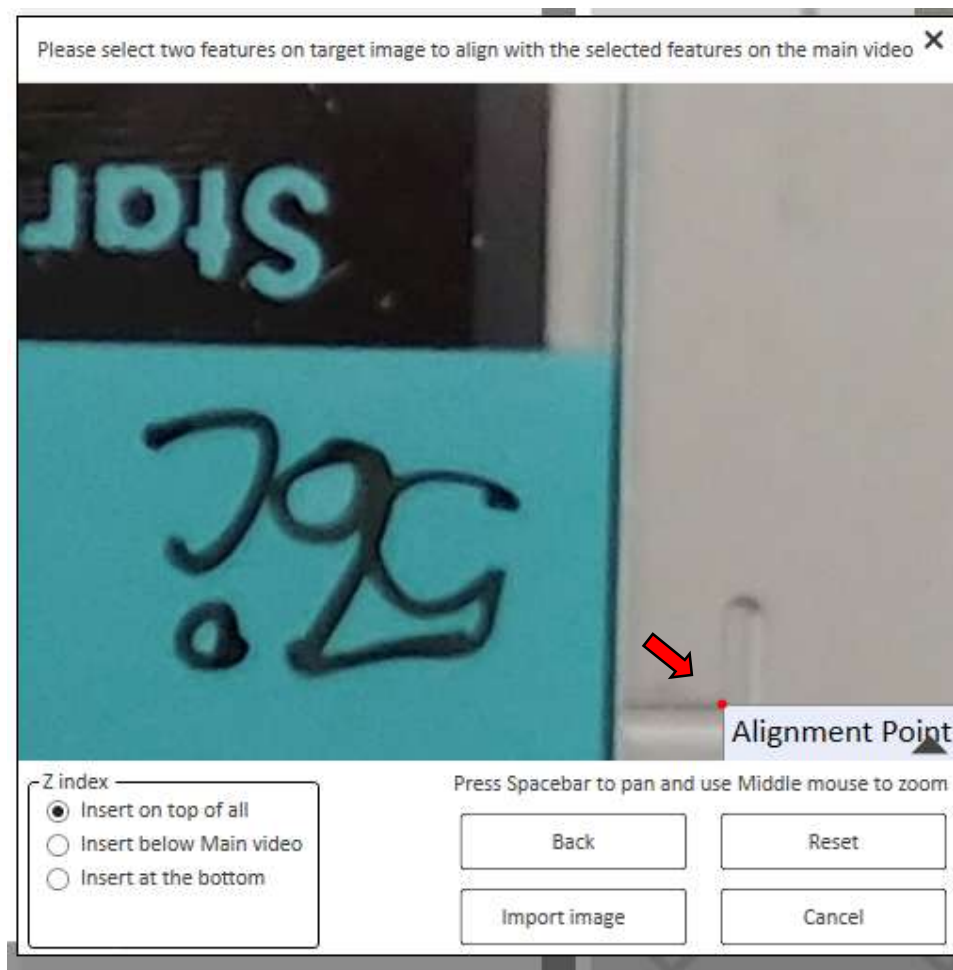
The fiducial interface can be moved to the side for better viewing of the camera feed. To set the fiducial, click the inner wedge of the fiducial.



Repeat the same steps for the bottom right fiducial, again selecting the inner edge of the fiducial.

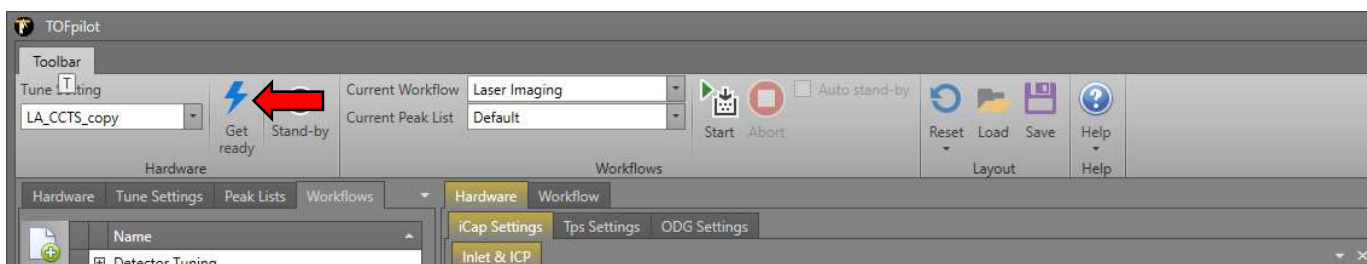


Close the inset menu by clicking the arrow in the upper right of the inset and set the fiducials on the imported image in the same order by clicking on the appropriate fiducials. The mouse wheel can be used to zoom in when mousing over a fiducial, and the reset button will resize the image. After setting both points, click the Import Image button.



Igniting the plasma

Once the purge sequence has finished (the bar under the purge button is completely green), select the Get Ready button in ToFPilot. Note: If the start button remains grayed out, restart the program.

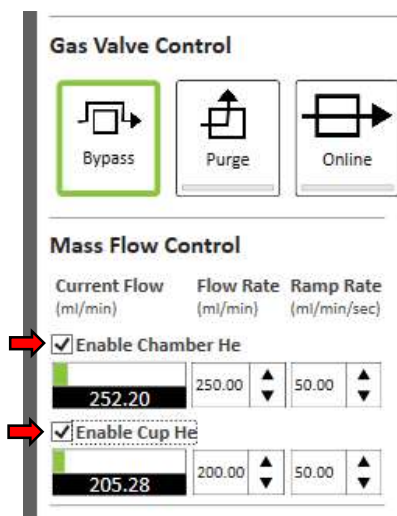


Remain close to the instrument while the plasma is being ignited since the ignition sequence can partially fail and melt the torch. **If the plasma sputters when igniting or is not a blue-green color, immediately (<1s) shut off the plasma by rotating the access port handle clockwise.** If the plasma needs to be shut off in this way, close the access port, click Standby, then simply click Get Ready again.



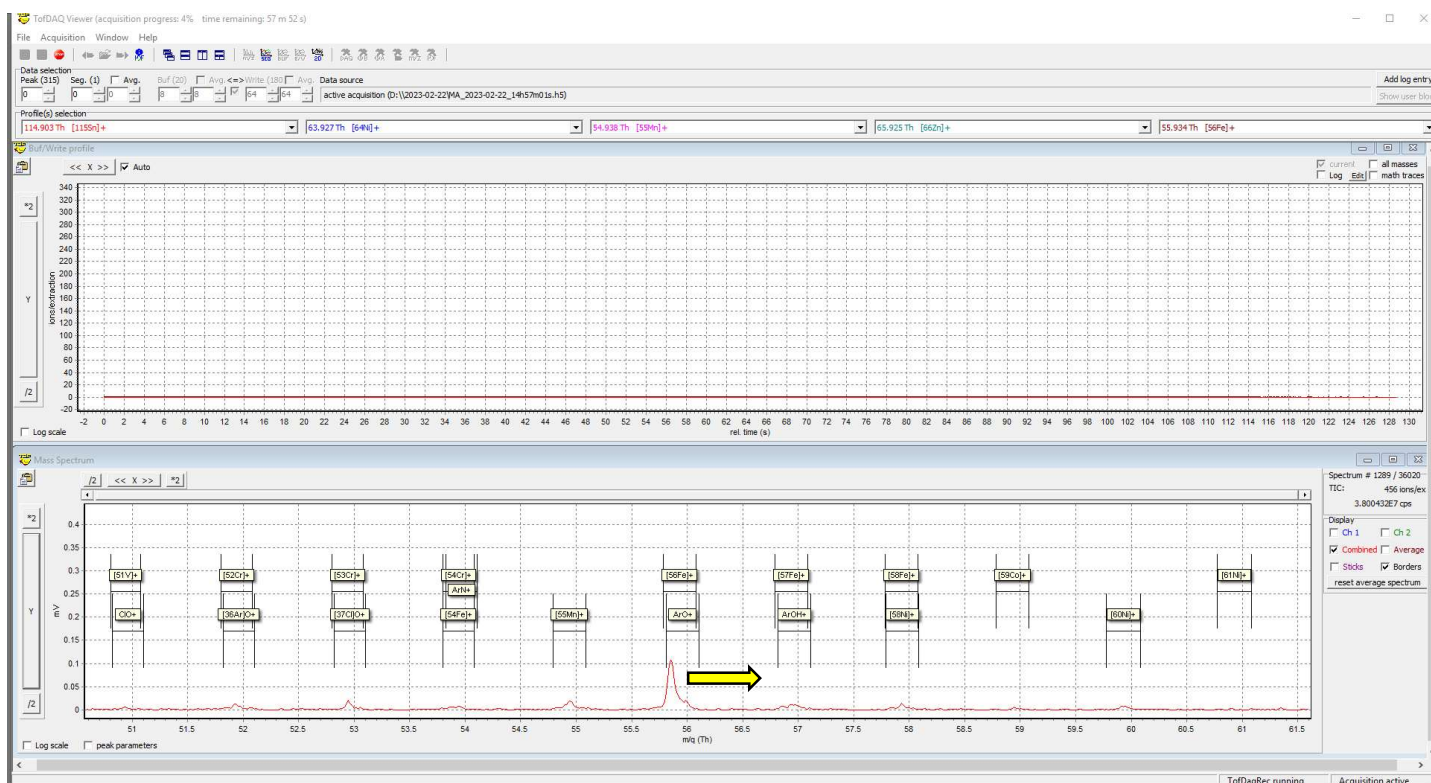
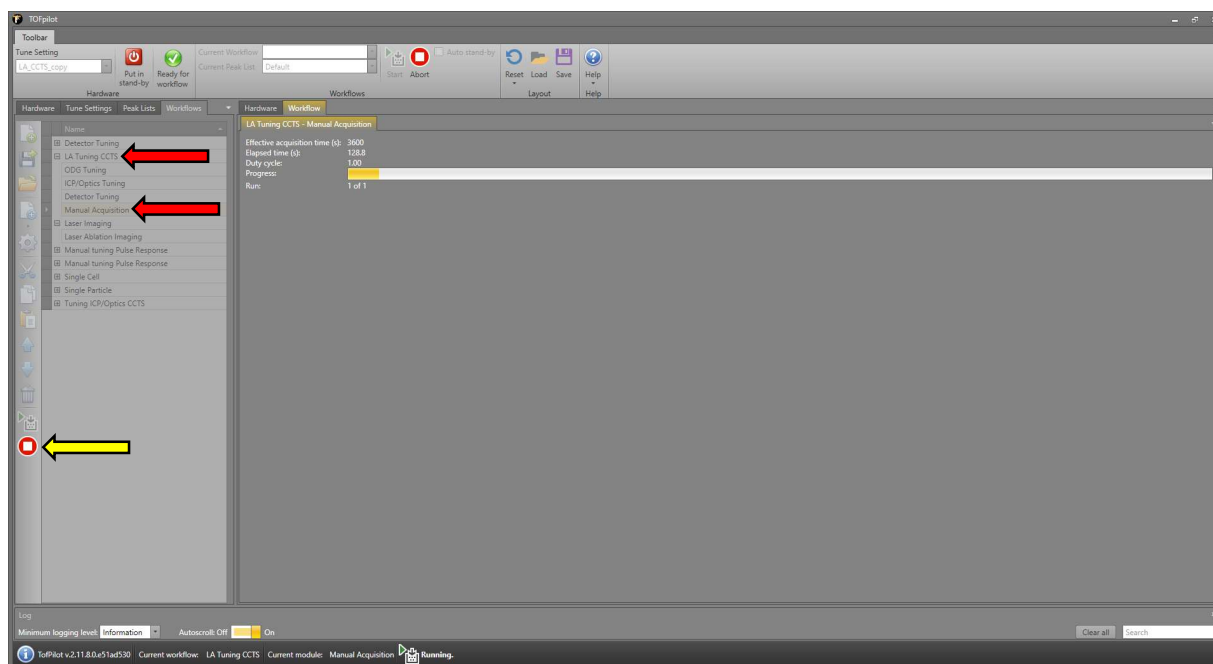
Turn on helium flow

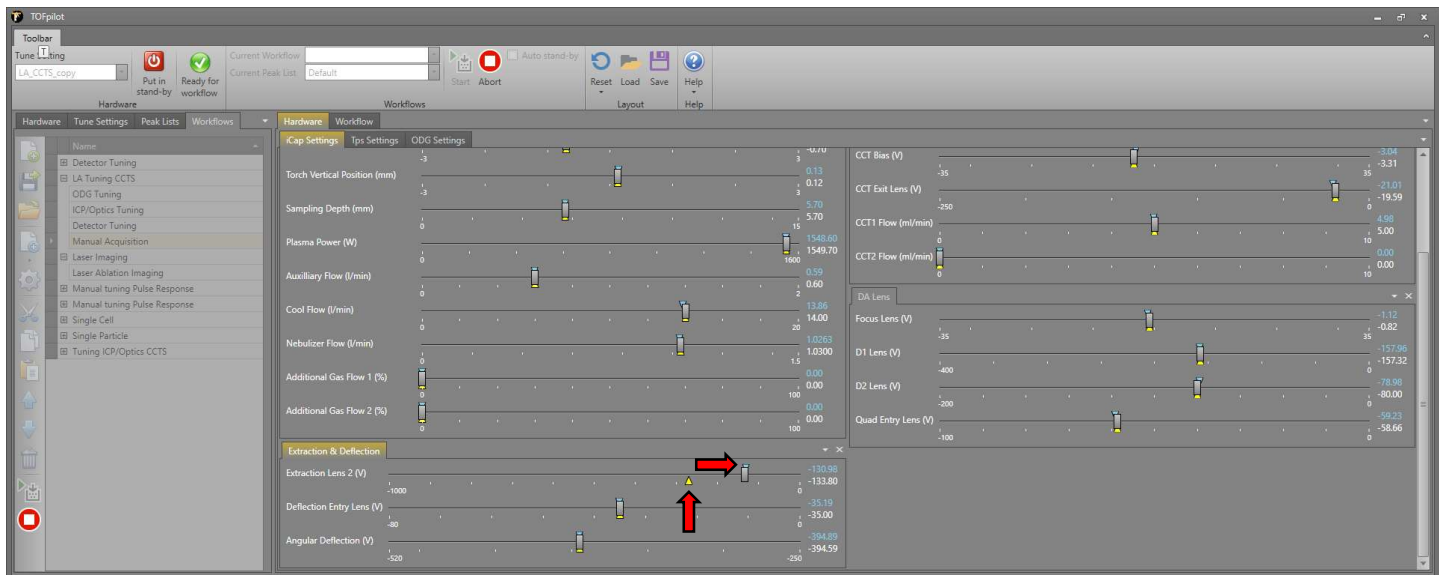
After the plasma has completed the ignition sequence, make sure that both the Chamber and Cup helium are activated by checking the respective boxes under Mass Flow Control. The Current Flow should roughly read the Flow Rate before starting an experiment.



Warming up the instrument

Under LA Tuning CCTS, select Manual Acquisition then click start to begin acquiring signal. Also turn up (make less negative) the extraction lens voltage. The peaks will slowly drift to the right as the electronics warm up – let this run for at least 15 minutes. To end the warm up, simply click Abort. Remember to return the extraction voltage to its original position after this step (click the yellow indicator to automatically set to the correct voltage).

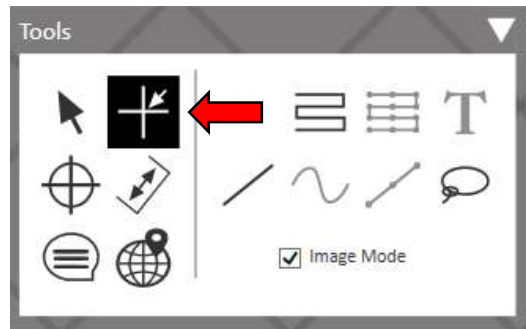




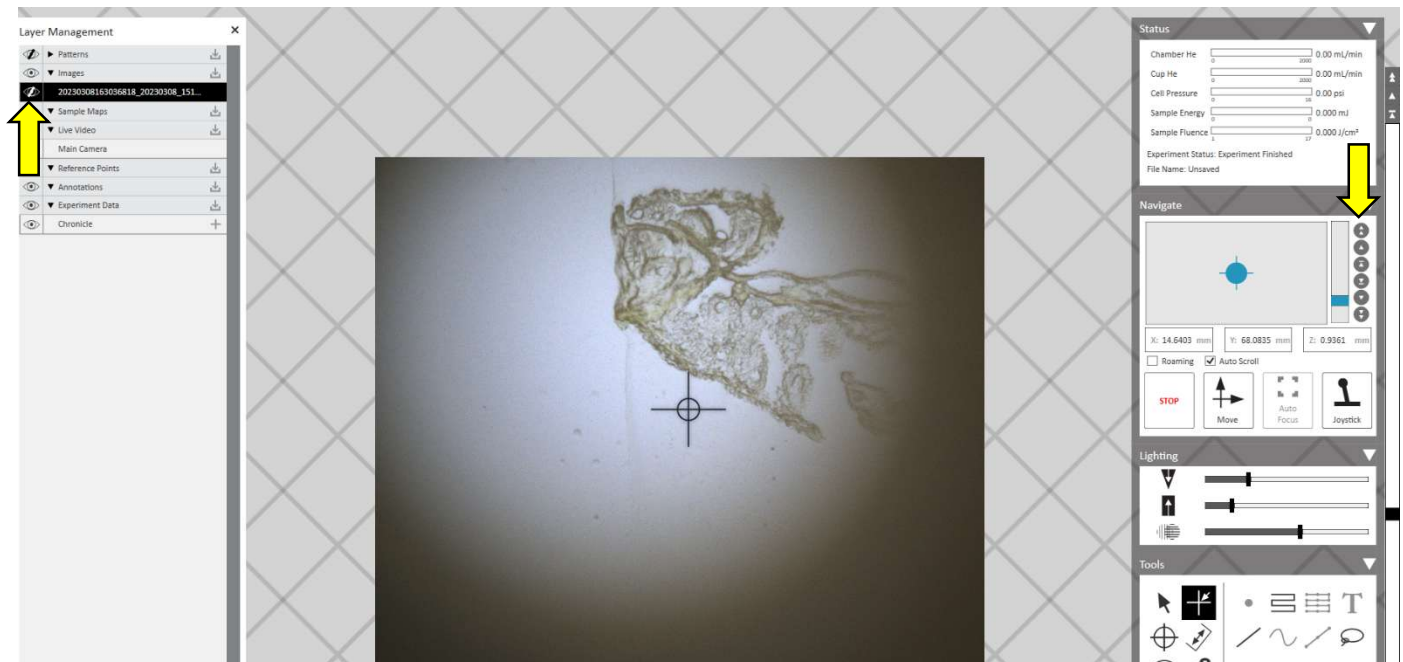
Scanning samples

Assigning ROIs

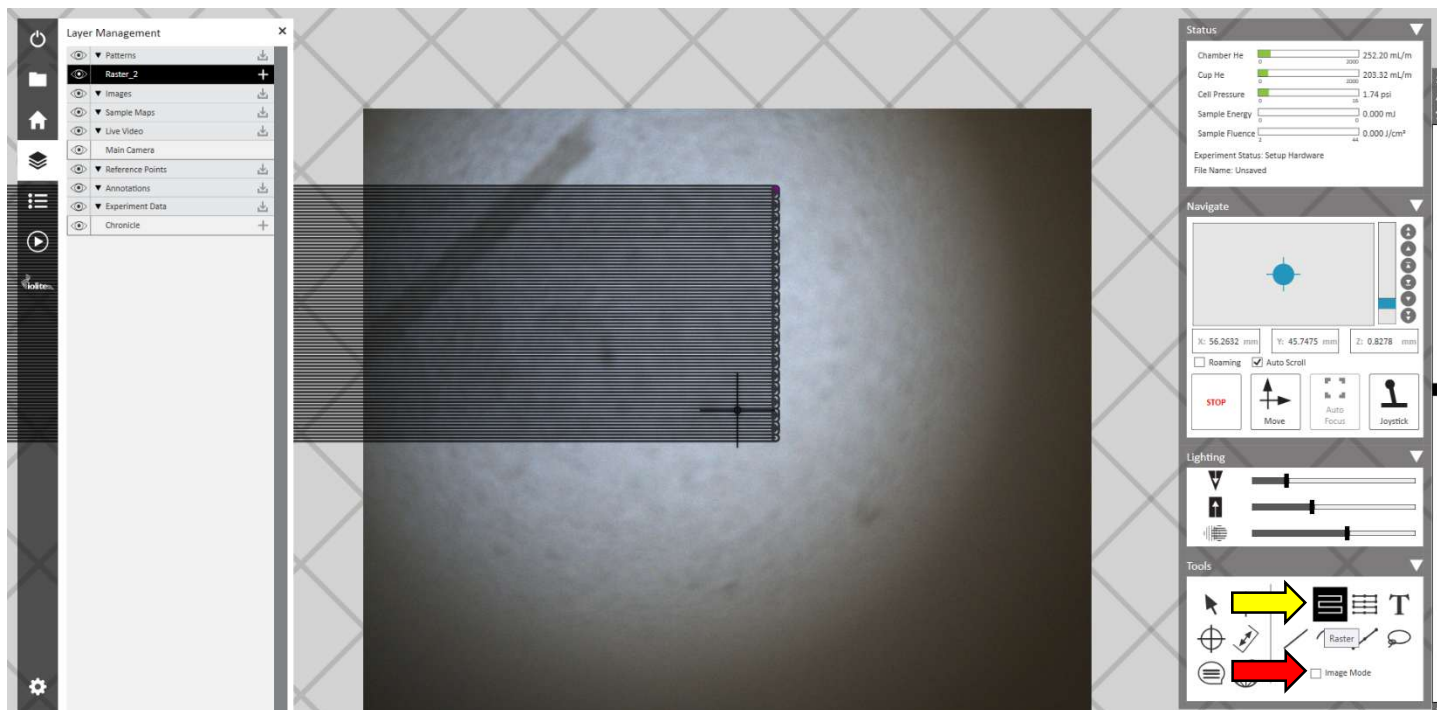
To navigate the stage in the laser software, use the Move To button and click on the screen to move to that point.



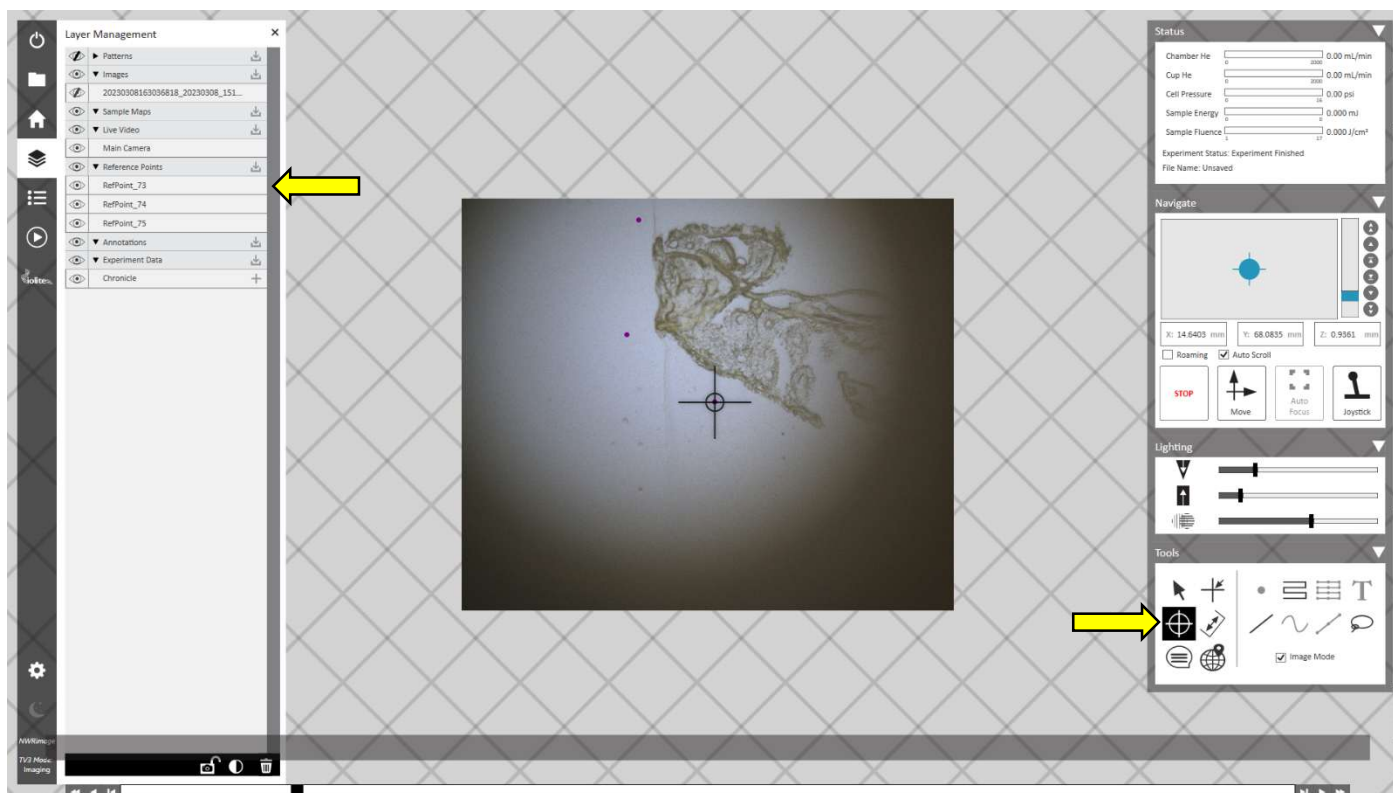
Clicking the eye next to the image will either show or hide the picture, which can be used to guide to the general area of the sample. The buttons on the right can be used to change the Z height to focus on the sample.



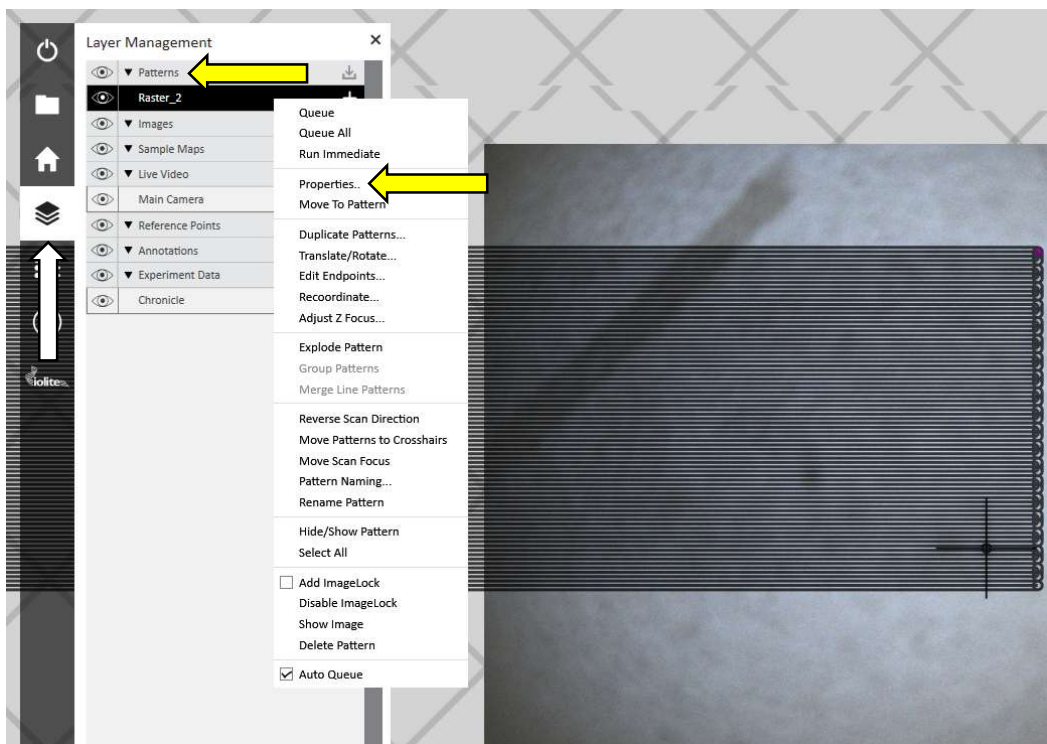
To assign a section to ablate, select the Image Line Raster mode then click to begin drawing an ROI. If the sample is larger than the field of view, simply drag the cursor to the edge of the screen to move the field of view. To finish drawing the ROI, click again. It is recommended that the raster area is slightly larger than the sample on all sides. Make sure “Image Mode” is checked before drawing the ROI.



Reference Points can also be assigned to assist in outlining the sample.



After drawing an ROI, select the Layers Management tab. The newly drawn raster will be under the Patterns section. Right click the pattern and select properties.



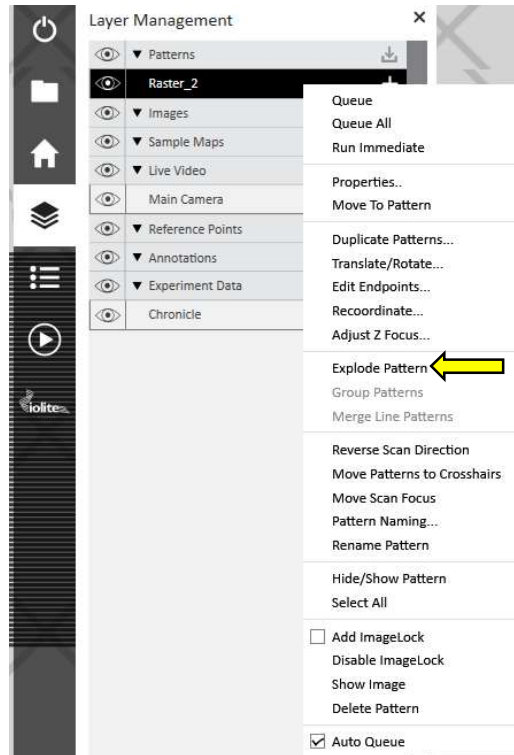
The properties tab allows for changing several aspects of the ablation throughout the experiment, however for general use the Energy (laser power), Rep Rate (scanning rate), IVA/XYR (spot size), and Raster Spacing (which should generally match the spot size) sections should be set based on the sample. It is recommended that the standards and samples are scanned with identical parameters if possible.

The screenshot displays the LACS software interface with the following settings:

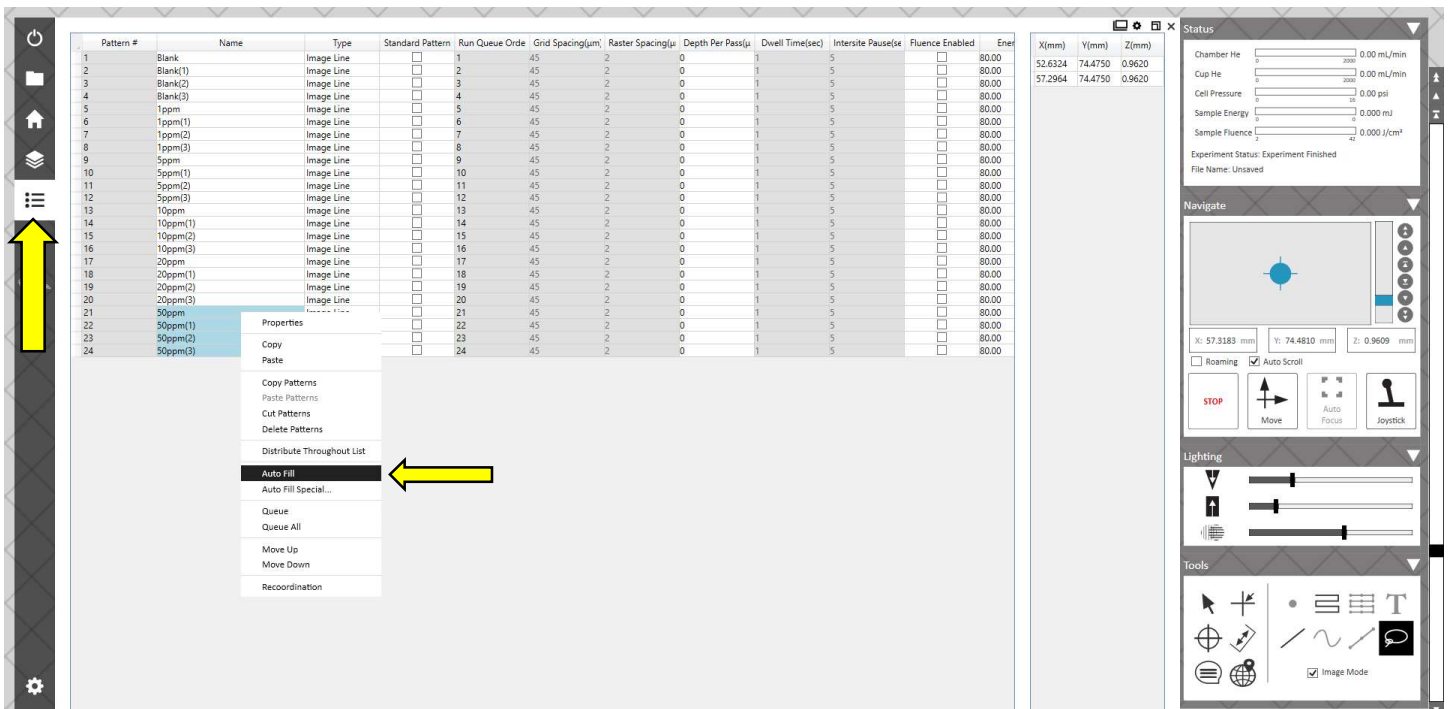
- Pattern Settings:**
 - Pattern Color: Blue
 - Name: Raster_2
 - Grid Spacing: [] []
 - Raster spacing: 20 [] []
 - ☒ Enable Ablation Pass
 - ☐ Enable Pre-Ablation Pass
 - ☐ Scan All Lines in Same Direction
 - ☒ Scan All Passes in Same Direction
- Ablation:**
 - Stage Settings:
 - Passes: 1 [] []
 - Scan Speed: 100 [] [] $\mu\text{m}/\text{sec}$
 - Depth per pass: 0 [] [] μm
 - Laser Settings:
 - Use Current Laser Settings
 - Apply Settings To Laser
 - Mode: 8x Firing []
 - Energy:
 - ☒ Percent
 - ☐ Fluence
 - Output: 100.00 [] [] %
 - Rep Rate: 10 [] [] Hz
 - IVA / XYR Shutter:
 - ☒ IVA
 - ☐ XYR
 - Slider: 20 [] [] μm
 - X: 0 [] [] μm
 - Y: 0 [] [] μm
 - R: 0 [] [] μm
 - ☐ Auto Rotate
 - Timing Settings:**
 - Dwell Time: [] [] sec
 - Burst Count: [] []
 - Intersite Pause: [] [] sec

Buttons at the bottom: Save As Default, Cancel, Ok.

Once the properties for a raster line are set, the pattern needs to be “exploded.” Right click the pattern, then click Explode Pattern to convert the ROI into raster lines.

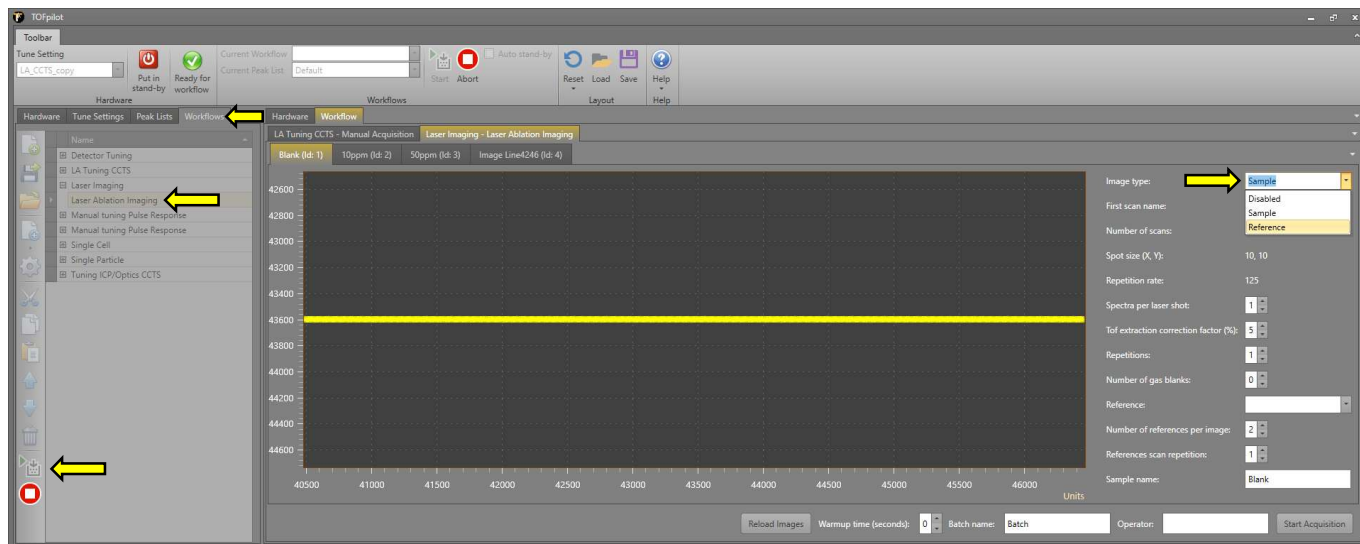


If the ROI is of a standard, click the Lines tab. Double click the first line of the set of newly exploded lines to rename it. After renaming the first line, highlight the set of lines, right click, and select Auto Fill. This will help in assigning standards to a sample later.



Initiating a scan

Once all of the ROIs have been assigned, under the Workflows tab in ToFPilot, select the Laser Imaging, then click Start. Each ROI will be brought up as a separate tab. For each standard, assign it as a Reference in the dropdown menu.



When selecting a sample, be sure to assign the appropriate standards to each sample. Also make sure the lines of the standards match what is assigned in the sample. If multiple samples ROIs are being run in one batch, the number of standard lines should be the sum of the lines of standards run per sample (i.e. 2 samples with 4 standard lines each means there should be 8 lines in a standard ROI). It is recommended to run at least one gas blank (currently the laser does not properly do a gas blank and can be omitted). The sample can be named at this step. Once the appropriate entries are filled in, click Start Acquisition to begin scanning the sample.

